AI Accelerator Program

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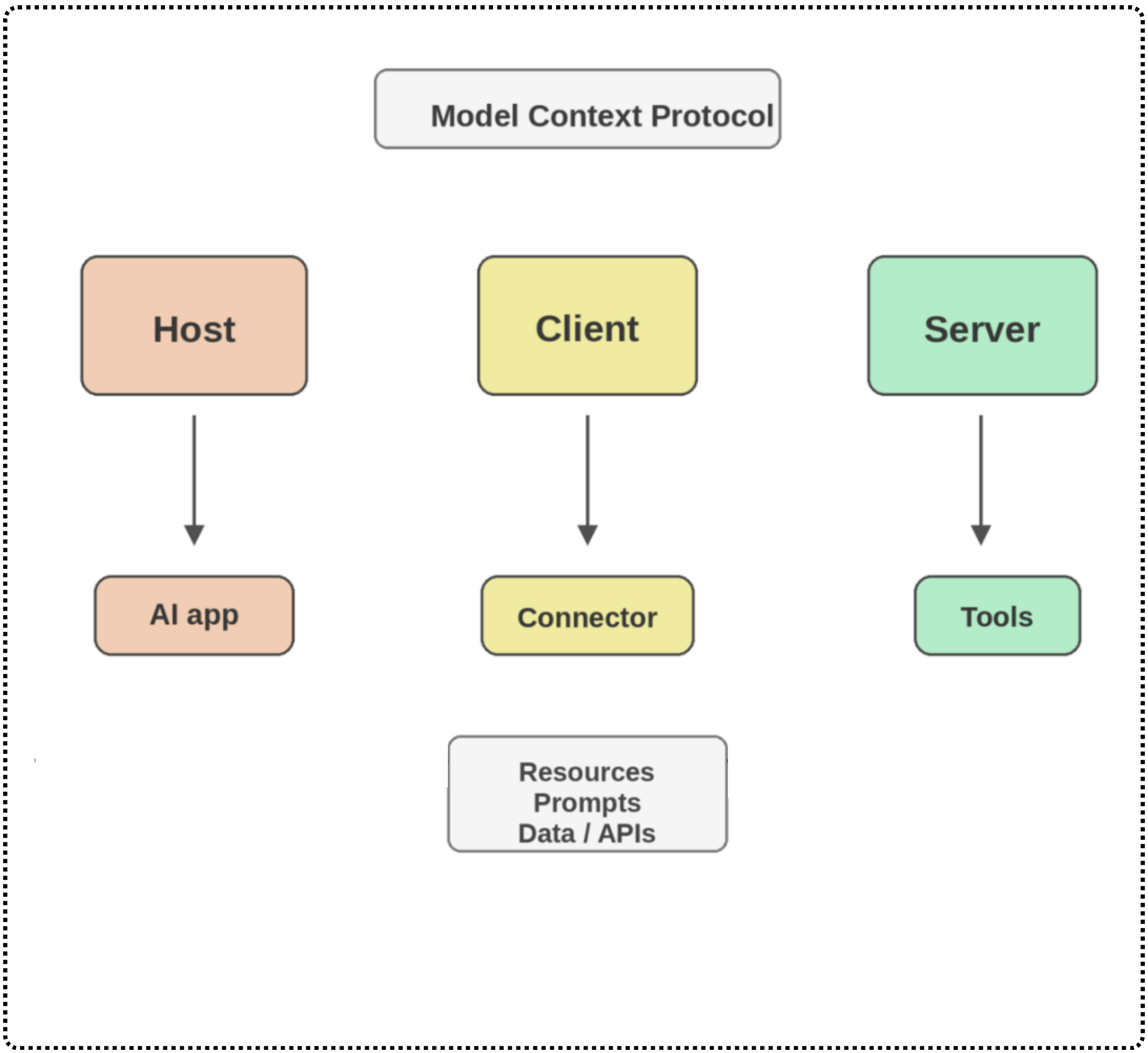
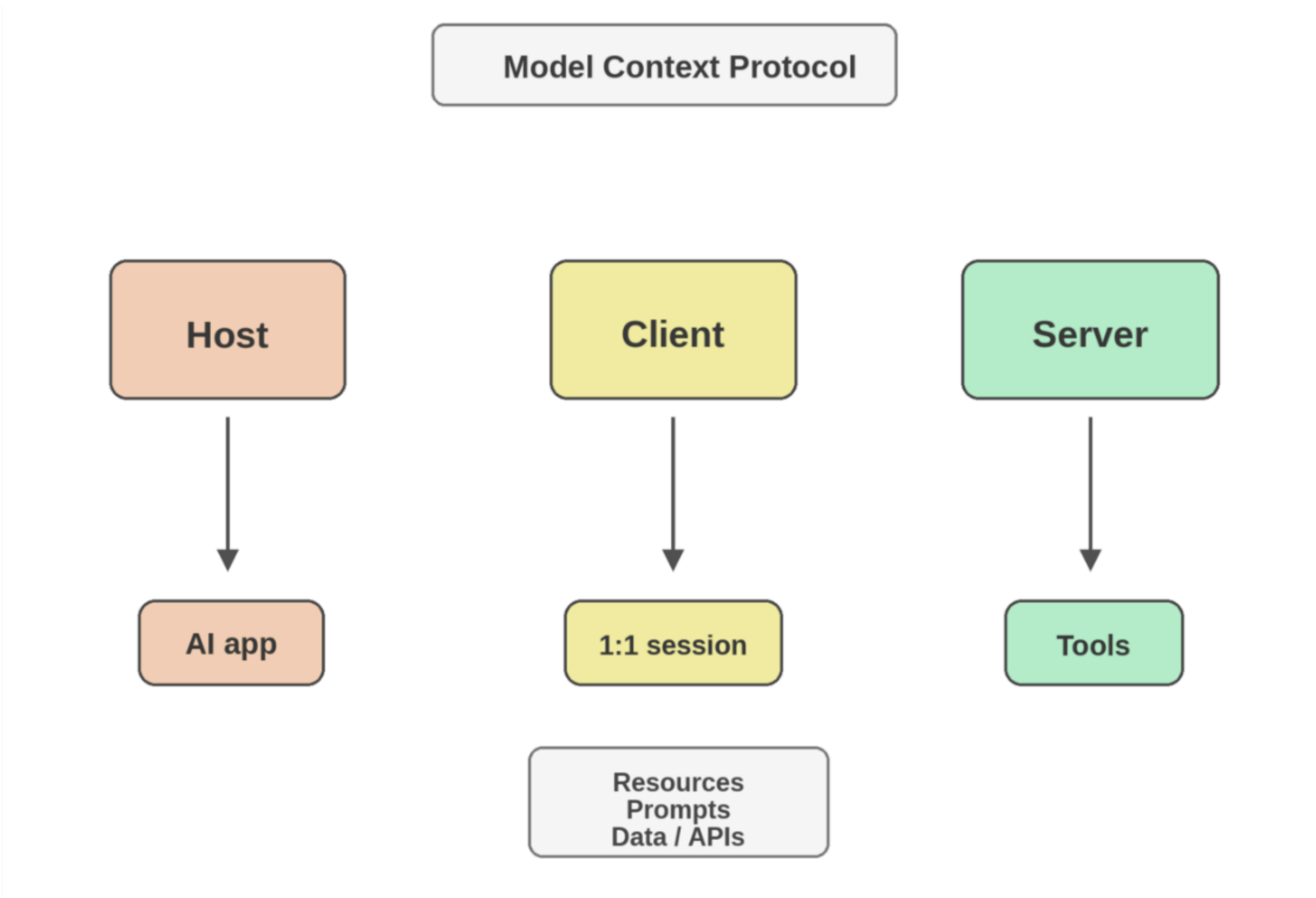
A Guide to Model Context Protocol(MCP)



What is MCP?

- Model Context Protocol (MCP) is a powerful open standard that was launched by Claude's parent company, Anthropic, in November 2024. It allows Large Language Models (LLMs) like Claude or GPT to seamlessly access external content such as documents, databases, tools, local files or APIs, in a structured, secure, and scalable way.



MCP at its core consists of three main components:

Model

It refers to the **LLM** (like Claude or GPT-5) that's responding to your inputs.

Context

It is the additional information the model needs to respond meaningfully. The source of this information can be documents, PDFs, prompts, or database entries.

Protocol

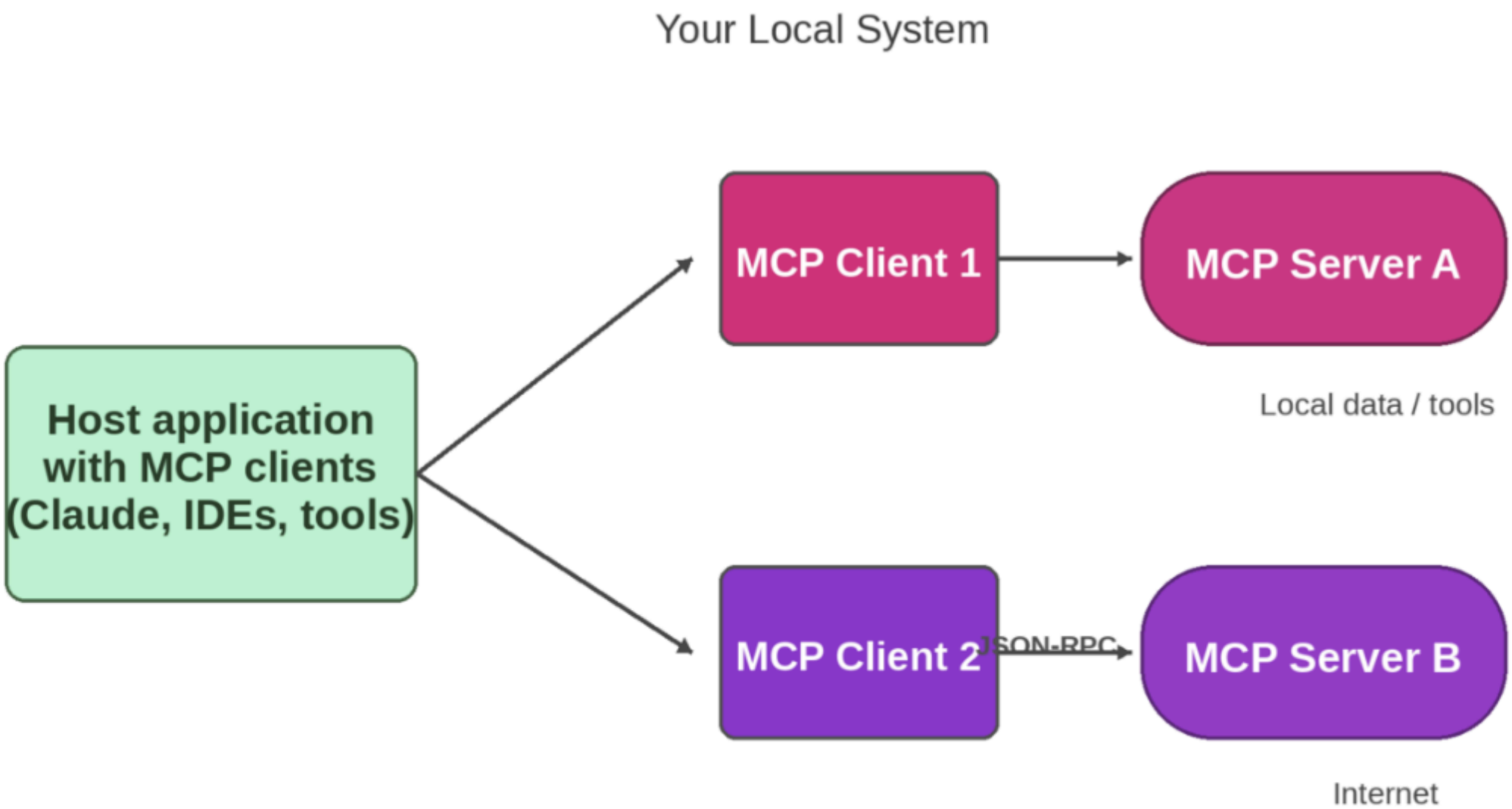
It represents a set of rules that lets the model access and use that context from structured sources.

How Does MCP Work?

In its true essence, MCP standardizes the communication between the clients and servers, without needing custom code or manual data uploads. Clients and Servers can mean different things for different technologies, but for MCP, these are:.

- **Host**: The app that manages the model, permissions, user interaction, and connected MCP clients.
- **Client**: A host-created connector that speaks MCP to one server over a stateful session.
- **Server**: The endpoint that exposes capabilities such as tools, resources, and prompt templates.

When you ask a question or give a command, the client sends a request to the server for the relevant information. The server then provides the necessary context (e.g., a file, email, or database), which the client uses to give you a response or complete a task. This process allows AI to efficiently work with real-time, dynamic data.



Thus, with MCP, the client (LLM interface) can automatically discover available servers (data sources), and request specific information based on your query.